

Deeraj Nagothu, Ph.D.

Hardware Systems Engineer, Microsoft.

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Professional Summary

Ph.D. Researcher specializing in production-level hardware and software systems: hardware validation, server architecture, and AI/ML deployment. 7+ years bridging hardware systems engineering with scalable ML solutions, from component-level design and validation to deploying LLM-powered agentic AI, transformer architectures, and distributed inference pipelines. Expertise spans electronic systems design, validation, and production AI frameworks for cloud/edge environments. Published 30+ papers while shipping production-ready code and open-source libraries. Applying deep technical expertise across the full hardware-software stack to advance system reliability and AI capabilities.

Technical Focus Areas

- ◇ LLM Development & Optimization: model training, transformer architectures, inference optimization, prompt engineering, and Agentic AI systems. Model evaluation, adversarial robustness & deepfake detection.
- ◇ Hardware Systems Engineering: Server architecture, component validation and testing, integrated circuitry design, hardware debugging, and system-level performance analysis.
- ◇ Computer Network security, Blockchain and Virtualization.

Work Experience

- Feb 2026 ◇ **Hardware Systems Engineer** Microsoft, Redmond, WA, USA.
- Aug'23 – Feb'26 ◇ **Research Scientist.** Intelligent Fusion Technology Inc. (IFT), Germantown, MD, USA.
 - Architected and deployed a transformer-based computer vision agentic pipeline integrating Stable Diffusion XL with Vision-Language Models (VLMs) for automated prompt engineering, improving YOLOv11 object detection recall by 33% on low-visibility targets.
 - Built and deployed real-time ML anomaly detection infrastructure processing data streams at scale, leveraging Elastic Stack for observability and Apache Airflow for orchestration. Implemented STIG-compliant security controls and monitoring dashboards, achieving 99.9% uptime.
 - Designed and implemented a neuro-symbolic Foundation model agent for automated hyperparameter tuning that leverages LLMs and VLMs with symbolic reasoning, achieving expert-level performance with 20% efficiency improvement on distributed extended Kalman filter optimization for multi-UAS positioning.
 - Designed a secure, blockchain access control and data integrity framework using containerized ML workflows (Docker/Kubernetes), and NDN architecture for encrypted data exchange.
- Jan'18 – Jul'23 ◇ **Research Assistant** SUNY Research Foundation, Binghamton University, Binghamton, NY, USA.
 - Designed and deployed a novel deepfake detection framework leveraging Electrical Network Frequency (ENF) signatures extracted from audio/video recordings, optimized for heterogeneous and low-power edge devices.
 - Built and open-sourced [pyenf](#), a Python library for multimodal ENF signal extraction and authentication with modular API design, with support for modular signal processing pipelines.
 - Integrated authentication system into blockchain-enabled smart surveillance platforms, ensuring verifiable data provenance in decentralized public safety infrastructures, and real-time media authentication using ENF as consensus.
 - Developed GAN fingerprinting techniques that leveraged frequency and artifact-based cues to detect AI-generated imagery, contributing to the early identification of synthetic media threats.

Skills

Deep Learning/LLMs	◇ Pytorch, Tensorflow, FastAI, Hugging Face Transformers, WandB, PEFT, Transformer-lens, NLTK, scikit-learn, Prompt Engineering, FastAPI.
Hardware Systems	◇ Server architecture and platform validation, Hardware debugging and troubleshooting, RCA, Signal integrity analysis, PCIe protocols, Memory subsystems (NUMA architecture), BIOS/firmware interfaces, Performance benchmarking.
Coding	◇ Python, C/C++, C#, Shell, Bash, MATLAB, Powershell, SQL, L ^A T _E X, Neo4j and Cypher.
Network and Security	◇ DHCP, SSH, VPN, DNS, Port-Forwarding, DMZ N/W, NMAP scan, Hyperledger Fabric, Pentesting using Metasploit framework, OpenDaylight for SDN, LibreNMS, Cacti Server, Tenable Nessus Scanner and DISA STIGs.
Virtualization	◇ ESXi, Vyos, Cisco Nexus, HyperV, Xen, OpenStack, OpenMano, Docker, Kubernetes, Proxmox, GCP and AWS platforms
Misc.	◇ Grafana, Kibana, Elasticsearch, Logstash, Apache Airflow, Wireshark, Git/GitHub.

Education

2017 – 2023	◇ Ph.D, Electrical & Computer Engineering , Binghamton University-SUNY, Binghamton, NY, USA. Dissertation title: <i>Lightweight Multimedia Authentication at the Edge using Environmental Fingerprint</i> Advisor: Dr. Yu Chen
2015 – 2016	◇ MS, Electrical & Computer Engineering , Binghamton University-SUNY, Binghamton, NY, USA. Thesis title: <i>iCrawl: A high interaction client honeypot system.</i> Advisor: Dr. Andrey Dolgikh
2011 – 2015	◇ B.Tech, Electronics & Communications Engineering , SASTRA University, Tamil Nadu, India.

Teaching Experience

Graduate Instructor	◇ Network Computer Security (EECE-480F),
Teaching Assistant	◇ Cyber Physical Systems (EECE-480A), Cryptography and Information Security (EECE-405/560), Linear Algebra and Engg Programming (EECE-212),

Research Publications

Journal Articles

- 1 Ogunbunmi, S., Chen, Y., Zhao, Q., **Nagothu, D.**, Wei, S., Chen, G., & Blasch, E. (2025). Interest Flooding Attacks in Named Data Networking and Mitigations: Recent Advances and Challenges. *Future Internet*, 17(8), 357. Publisher: Multidisciplinary Digital Publishing Institute.  doi:10.3390/fi17080357
- 2 Zhao, Q., **Nagothu, D.**, Tian, X., Chen, G., Pham, K. D., & Blasch, E. (2025). Sd-sat: Software-defined multi-constellation satellite communication traffic management framework. *IET Conference Proceedings*, 2024(31), 71–78. Publisher: The Institution of Engineering and Technology.  doi:10.1049/icp.2024.4615
- 3 Xu, R., **Nagothu, D.**, Chen, Y., Aved, A., Ardiles-Cruz, E., & Blasch, E. (2024). A Secure Interconnected Autonomous System Architecture for Multi-Domain IoT Ecosystems. *IEEE Communications Magazine*, 62(7), 52–57. Conference Name: IEEE Communications Magazine.  doi:10.1109/MCOM.001.2300354
- 4 Qu, Q., Hatami, M., Xu, R., **Nagothu, D.**, Chen, Y., Li, X., ... Chen, G. (2024). The microverse: A task-oriented edge-scale metaverse. *Future Internet*, "16"(2), 60.  doi:10.3390/fi16020060
- 5 **Nagothu, D.**, Xu, R., Chen, Y., Blasch, E., & Aved, A. (2022a). Defakepro: Decentralized deepfake attacks detection using enf authentication. *IT Professional*, 24(5), 46–52.  doi:10.1109/MITP.2022.3172653
- 6 **Nagothu, D.**, Xu, R., Chen, Y., Blasch, E., & Aved, A. (2022b). Deterring deepfake attacks with an electrical network frequency fingerprints approach. *Future Internet*, 14(5), 125.  doi:10.3390/fi14050125

^oThe complete list of publications is available on my [Google Scholar](#) page

- 7 Xu, R., **Nagothu, D.**, & Chen, Y. (2021a). Decentralized video input authentication as an edge service for smart cities. *IEEE Consumer Electronics Magazine*, 10(6), 76–82. [doi:10.1109/MCE.2021.3062564](https://doi.org/10.1109/MCE.2021.3062564)
- 8 Xu, R., **Nagothu, D.**, & Chen, Y. (2021b). Econledger: A proof-of-enf consensus based lightweight distributed ledger for iomt networks. *Future Internet*, 13(10), 248. [doi:10.3390/fi13100248](https://doi.org/10.3390/fi13100248)
- 9 **Nagothu, D.**, Chen, Y., Aved, A., & Blasch, E. (2021). Authenticating video feeds using electric network frequency estimation at the edge. *EAI Endorsed Transactions on Security and Safety*, "7"(24). [doi:10.4108/eai.4-2-2021.168648](https://doi.org/10.4108/eai.4-2-2021.168648)
- 10 Xu, R., Nikouei, S. Y., **Nagothu, D.**, Fitwi, A., & Chen, Y. (2020). Blendsps: A blockchain-enabled decentralized smart public safety system. *Smart Cities*, 3(3), 928–951. [doi:10.3390/smartcities3030047](https://doi.org/10.3390/smartcities3030047)
- 11 **Nagothu, D.**, Chen, Y., Blasch, E., Aved, A., & Zhu, S. (2019). Detecting malicious false frame injection attacks on surveillance systems at the edge using electrical network frequency signals. *Sensors (Basel)*, 19(11), 1–19. [doi:10.3390/s19112424](https://doi.org/10.3390/s19112424)

Conference Proceedings

- 1 Hatami, M., Qu, Q., **Nagothu, D.**, Mohammadi, J., Chen, Y., Ardiles-Cruz, E., & Blasch, E. (2025). Securing Smart Grid Digital Twins via Real-World ENF Anchors Against Deepfake Attacks. (pp. 392–397). [doi:10.1109/PerComWorkshops65533.2025.00097](https://doi.org/10.1109/PerComWorkshops65533.2025.00097)
- 2 **Nagothu, D.**, Bao, Y., Zhuang, P., Zhang, Y., Chen, G., Blasch, E., & Pham, K. (2025). An automated framework for enhancing ai systems through targeted synthetic data generation. In *Naecon 2025-ieee national aerospace and electronics conference* (pp. 1–5). IEEE.
- 3 Qu, Q., Sun, H., Chen, J., **Nagothu, D.**, & Chen, Y. (2025). Secure avatars via environmental fingerprints for virtual health monitoring services. In *Icc 2025-ieee international conference on communications* (pp. 4915–4920). IEEE.
- 4 Xu, R., Liu, X., **Nagothu, D.**, Qu, Q., & Chen, Y. (2025). Detecting Manipulated Digital Entities Through Real-World Anchors. In L. Barolli (Ed.), *Advanced Information Networking and Applications* (pp. 450–461). [doi:10.1007/978-3-031-87784-1_41](https://doi.org/10.1007/978-3-031-87784-1_41)
- 5 Pazytkarim, A., **Nagothu, D.**, & Chen, Y. (2024). A lightweight deep learning model for rapid detection of fabricated ENF signals from audio sources. In *Disruptive Technologies in Information Sciences VIII* (Vol. 13058, pp. 363–375). [doi:10.1117/12.3013456](https://doi.org/10.1117/12.3013456)
- 6 Poredi, N., Sudarsan, M., Solomon, E., **Nagothu, D.**, & Chen, Y. (2024). Generative adversarial networks-based AI-generated imagery authentication using frequency domain analysis. In *Disruptive Technologies in Information Sciences VIII* (Vol. 13058, pp. 376–390). [doi:10.1117/12.3013240](https://doi.org/10.1117/12.3013240)
- 7 Zhao, X., **Nagothu, D.**, & Chen, Y. (2024). A homogeneous low-resolution face recognition method using correlation features at the edge. In *Sensors and Systems for Space Applications XVII* (Vol. 13062, pp. 107–123). [doi:10.1117/12.3008368](https://doi.org/10.1117/12.3008368)
- 8 Poredi, N., **Nagothu, D.**, & Chen, Y. (2024). Authenticating ai-generated social media images using frequency domain analysis. In *2024 ieee 21st consumer communications & networking conference (ccnc)* (pp. 534–539). ISSN: 2331-9860. [doi:10.1109/CCNC51664.2024.10454640](https://doi.org/10.1109/CCNC51664.2024.10454640)
- 9 **Nagothu, D.**, Xu, R., Chen, Y., Blasch, E., & Ardiles-Cruz, E. (2023). Application of Electrical Network Frequency as an Entropy Generator in Distributed Systems. In *NAECON 2023 - IEEE National Aerospace and Electronics Conference* (pp. 233–238). ISSN: 2379-2027. [doi:10.1109/NAECON58068.2023.10365792](https://doi.org/10.1109/NAECON58068.2023.10365792)
- 10 Parker, J., **Nagothu, D.**, & Chen, Y. (2023). Decentralized Vehicular Identification and Tracking on Lightweight IoT Edge Nodes. In *NAECON 2023 - IEEE National Aerospace and Electronics Conference* (pp. 198–203). ISSN: 2379-2027. [doi:10.1109/NAECON58068.2023.10365952](https://doi.org/10.1109/NAECON58068.2023.10365952)
- 11 **Nagothu, D.**, Xu, R., & Chen, Y. (2023). DEMA: Decentralized electrical network frequency map for social media authentication. In *Disruptive Technologies in Information Sciences VII* (Vol. 12542, pp. 57–72). [doi:10.1117/12.2663303](https://doi.org/10.1117/12.2663303)
- 12 Poredi, N., **Nagothu, D.**, & Chen, Y. (2023). AUSOME: Authenticating social media images using frequency analysis. In *Disruptive Technologies in Information Sciences VII* (Vol. 12542, pp. 44–56). [doi:10.1117/12.2663296](https://doi.org/10.1117/12.2663296)

- 13 Poredi, N., **Nagothu, D.**, Chen, Y., Li, X., Aved, A., Ardiles-Cruz, E., & Blasch, E. (2022). Robustness of electrical network frequency signals as a fingerprint for digital media authentication. In *2022 IEEE 24th International Workshop on Multimedia Signal Processing (MMSP)* (pp. 1–6). [doi:10.1109/MMSP55362.2022.9949315](https://doi.org/10.1109/MMSP55362.2022.9949315)
- 14 **Nagothu, D.**, Dimock, D., Kulesza, A., Yang, H., & Chen, Y. (2022). A distributed crawler for iomt-based public safety surveillance exploring the spatio-temporal correlation. In *Sensors and systems for space applications xv* (Vol. 12121, pp. 18–28). [doi:10.1117/12.2618909](https://doi.org/10.1117/12.2618909)
- 15 **Nagothu, D.**, Xu, R., Chen, Y., Blasch, E., & Aved, A. (2021a). Detecting compromised edge smart cameras using lightweight environmental fingerprint consensus. In *Proceedings of the 19th ACM conference on embedded networked sensor systems* (pp. 505–510). [doi:10.1145/3485730.3493684](https://doi.org/10.1145/3485730.3493684)
- 16 **Nagothu, D.**, Xu, R., Chen, Y., Blasch, E., & Aved, A. (2021b). Defake: Decentralized enf-consensus based deepfake detection in video conferencing. In *IEEE 23rd International Workshop on Multimedia Signal Processing*. [doi:10.1109/MMSP53017.2021.9733503](https://doi.org/10.1109/MMSP53017.2021.9733503)
- 17 Quan, W., **Nagothu, D.**, Poredi, N., & Chen, Y. (2021). Cripi: An efficient critical pixels identification algorithm for fast one-pixel attacks. In *Sensors and systems for space applications xiv* (Vol. 11755, pp. 83–99). [doi:10.1117/12.2581377](https://doi.org/10.1117/12.2581377)
- 18 Rosenberg, M., Burns, J. H., **Nagothu, D.**, & Chen, Y. (2020). Enabling continuous operations for uavs with an autonomous service network infrastructure. In *Sensors and systems for space applications xiii* (Vol. 11422, pp. 165–179). [doi:10.1117/12.2565866](https://doi.org/10.1117/12.2565866)
- 19 Fitwi, A. H., **Nagothu, D.**, Chen, Y., & Blasch, E. (2019). A distributed agent-based framework for a constellation of drones in a military operation. In *Proc. - winter simul. conf.* (Vol. 2019-Decem). [doi:10.1109/WSC40007.2019.9004907](https://doi.org/10.1109/WSC40007.2019.9004907)
- 20 **Nagothu, D.**, Schwell, J., Chen, Y., Blasch, E., & Zhu, S. (2019). A study on smart online frame forging attacks against video surveillance system. In *Proc. SPIE - Int. Soc. Opt. Eng.* (Vol. 11017). [doi:10.1117/12.2519005](https://doi.org/10.1117/12.2519005)
- 21 **Nagothu, D.**, Xu, R., Nikouei, S. Y., & Chen, Y. (2019). A microservice-enabled architecture for smart surveillance using blockchain technology. In *2018 IEEE Int. Smart Cities Conf. ISC2 2018*. [doi:10.1109/ISC2.2018.8656968](https://doi.org/10.1109/ISC2.2018.8656968)
- 22 Nikouei, S. Y., Xu, R., **Nagothu, D.**, Chen, Y., Aved, A., & Blasch, E. (2019). Real-time index authentication for event-oriented surveillance video query using blockchain. In *2018 IEEE Int. Smart Cities Conf. ISC2 2018*. [doi:10.1109/ISC2.2018.8656668](https://doi.org/10.1109/ISC2.2018.8656668)
- 23 **Nagothu, D.**, & Dolgikh, A. (2017). Icrawl: A visual high interaction web crawler. In *Lect. Notes Comput. Sci. (including subser. Lect. Notes Artif. Intell. Lect. Notes Bioinformatics)* (Vol. 10446 LNCS). [doi:10.1007/978-3-319-65127-9_8](https://doi.org/10.1007/978-3-319-65127-9_8)

Book Chapters

- 1 Xu, R., **Nagothu, D.**, Chen, Y., Xu, R., **Nagothu, D.**, & Chen, Y. (2024). AR-Edge: Autonomous and Resilient Edge Computing Architecture for Smart Cities. [doi:10.5772/intechopen.1005876](https://doi.org/10.5772/intechopen.1005876)
- 2 **Nagothu, D.**, Poredi, N., & Chen, Y. (2022). Evolution of attacks on intelligent surveillance systems and effective detection techniques. [doi:10.5772/intechopen.105958](https://doi.org/10.5772/intechopen.105958)
- 3 Xu, R., **Nagothu, D.**, & Chen, Y. (2022). Ecom: Epoch randomness-based consensus committee configuration for IoT blockchains. In K. Daimi, I. Dionysiou, & N. El Madhoun (Eds.), *Principles and practice of blockchains* (pp. 135–154). [doi:10.1007/978-3-031-10507-4_7](https://doi.org/10.1007/978-3-031-10507-4_7)
- 4 **Nagothu, D.**, Xu, R., Nikouei, S. Y., Zhao, X., & Chen, Y. (2020). Smart surveillance for public safety enabled by edge computing. (pp. 409–433). [doi:10.1049/PBPC033E_ch19](https://doi.org/10.1049/PBPC033E_ch19)

Books

- 1 **Nagothu, D.**, & Chen, Y. (2023). *Authentication of video feeds in smart edge surveillance networks* (C. Olson, Ed.). Bellingham, Washington 98227-0010: SPIE Press.

Dissertation and Thesis

- 1 **Nagothu, D.** (2023). *Lightweight Multimedia Authentication at the Edge Using Environmental Fingerprint* (Ph.D. State University of New York at Binghamton, United States – New York). ISBN: 9798380566988. Retrieved October 28, 2023, from <https://www.proquest.com/docview/2872097834/abstract/A49AAD7CD800446BPQ/1>
- 2 **Nagothu, D.** (2016). *Icrawl: A high interaction client honeypot system* (M.S. State University of New York at Binghamton, United States – New York).

Professional Activities

Reviewer for Journals

- ◇ IEEE Transactions on Pattern Analysis and Machine Intelligence
- ◇ IEEE Transactions on Multimedia Computing Communications
- ◇ IEEE Transactions on Dependable and Services Computing
- ◇ IEEE Transactions on Services Computing
- ◇ IEEE Transactions on Networking
- ◇ IEEE Internet of Things Journal
- ◇ SPIE Journal of Electronic Imaging (JEI).
- ◇ Elsevier Computers and Security
- ◇ Expert Systems with Applications
- ◇ IEEE Access
- ◇ IEEE Transactions on Aerospace and Electronic Systems
- ◇ IEEE Transactions on Cloud Computing
- ◇ IEEE Computers
- ◇ Applied Sciences
- ◇ MDPI Sensors

Reviewer for Conferences

- ◇ IEEE International Conference on Computer Communications (INFOCOM).
- ◇ IEEE Communications Magazine (COMMAG)
- ◇ IEEE Signal Processing Letters
- ◇ IEEE Global Communications Conference (GLOBECOM) IoT and Sensor Networks (IoTSN).
- ◇ IEEE Global Communications Conference (GLOBECOM) Communication and Information Systems Security (CISS).
- ◇ IEEE Global Communications Conference (GLOBECOM) Communications Software, Services and Multimedia Apps (CSSMA).
- ◇ IEEE International Conference on Wireless and Mobile Computing, Networking And Communications (WiMob).
- ◇ ACM International Workshop on Blockchain-enabled Networked Sensor Systems (BlockSys)
- ◇ IEEE International Smart Cities Conference (ISC2).
- ◇ IEEE International Conference on Cloud Networking (CloudNet)
- ◇ IEEE International Conference on Communications (ICC)
- ◇ Knowledge based Systems (KNOSYS)
- ◇ Scientific Reports

Miscellaneous Experience

Awards and Achievements

- 2025 ◇ **Best Paper Award**, Best Workshop Paper at IEEE Percom SPT-IoT Conference. (Securing Smart Grid Digital Twins via Real-World ENF Anchors against DeepFake Attacks)

Miscellaneous Experience (continued)

- ◇ **Finalist**, xTech AI Competition, for the project "GRADIENT", focused on synthetic data-based feature engineering for robust AI model performance.
- ◇ **Winner** IEEE AESS Cybersecurity Challenge: Authored the winning challenge proposal "Federated AI for Resilient Avionics and Drone Operations," establishing a federated learning benchmark for resilient aerospace and drone cybersecurity research.
- 2020 ◇ **GSEA**, Graduate Student Award for Excellence in Teaching (**Courses** - Network Computer Security and Cyber Physical Systems).